

Document Control

- Document Title: Business Requirements Document (BRD)
- System Name: Application

Executive Summary

- This document defines the business requirements for the Application system.
- The application ingests customer and order CSV data into Delta tables, performs data cleansing, computes TotalAmount, creates ordersummary with SCD Type 2 handling, and aggregates daily spend into customeraggregatespend.
- The process is implemented via Delta Live Tables.

Business Objectives

- Load source CSVs into Delta tables for customer and order data.
- Cleanse data by removing "Null"/Null literals and duplicates.
- Enrich orders with a computed TotalAmount = PricePerUnit × Qty.
- Produce an ordersummary table by joining customer and order on CustId.
- Apply SCD Type 2 change handling for customer data.
- Create a customeraggregatespend table aggregating spend per Name per Date.
- Execute end-to-end pipeline using Delta Live Tables.

In-Scope

- Source ingestion from specified file system locations.
- Creation and management of Delta tables: customer, order, ordersummary, customeraggregatespend.
- Data cleansing actions as stated.
- Computation of TotalAmount in order.
- Join logic and SCD Type 2 handling for customer changes.
- Aggregate processing from ordersummary to customeraggregatespend.
- Implementation using Delta Live Tables.

Out-of-Scope

- Any data sources, systems, or fields not present in requirements.
- Security, access controls, SLAs, monitoring, alerting.
- Performance tuning and scaling.
- Scheduling, orchestration, or DevOps beyond "Implement steps 1-10 using Delta Live Tables."
- Data quality metrics beyond cleansing actions listed.
- User interfaces, reports, or analytics consumption layers.

Stakeholders

- Information not present in requirements.

Business Requirements

BR-CORE-001 — Source Data Ingestion

- Description: Load source customer and order CSVs into Delta tables from specified file paths.
- Business Value: Enables structured ingestion of raw business data for downstream processing.
- Priority: High

BR-CORE-002 — Schema Management

- Description: Maintain schemas as defined for customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId).
- Business Value: Ensures consistent data structure for processing and reporting.
- Priority: High

BR-CORE-003 — Computed Column (TotalAmount)

- Description: Add computed column TotalAmount on order table as $\text{PricePerUnit} \times \text{Qty}$.
- Business Value: Captures essential transaction value for analytics and aggregation.
- Priority: High

BR-CORE-004 — Data Cleansing

- Description: Remove "Null"/Null literals and duplicates in customer and order data.
- Business Value: Improves data quality for reliable processing and reporting.
- Priority: High

BR-CORE-005 — Ordersummary Table Creation

- Description: Create ordersummary table via join of customer and order on CustId, applying SCD Type 2 for changes.
- Business Value: Enables full historical tracking and integrated view of customer orders.
- Priority: High

BR-CORE-006 — SCD Type 2 Handling

- Description: Handle customer changes using SCD Type 2 rules—close old records, open new, set Active/Inactive, manage StartDate/EndDate.
- Business Value: Permits accurate historical lineage and data traceability.
- Priority: High

BR-CORE-007 — Ordersummary Schema Enforcement

- Description: Ensure ordersummary includes CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date.
- Business Value: Maintains analytic readiness and join traceability.
- Priority: High

BR-CORE-008 — Aggregate Spend Calculation

- Description: Create customeraggregatespend table by aggregating TotalAmount per Name and Date.
- Business Value: Supports business analysis of customer spending patterns.
- Priority: High

BR-CORE-009 — End-to-End Pipeline Implementation

- Description: Implement all functional steps using Delta Live Tables.
- Business Value: Ensures scalable, automated, production-grade deployment.
- Priority: High

Assumptions & Constraints

- All data sources and targets are limited to those specified.
- Attributes for SCD Type 2 (StartDate, EndDate, Active) are referenced but not fully defined.
- Implementation is constrained to usage of Delta Live Tables as the orchestration solution.

Success Metrics

- 100% of ingested records are loaded according to defined schemas.
- All duplicate and "Null" records are removed as per rules.
- Ordersummary and customeraggregatespend tables reflect accurate joins, SCD Type 2 handling, and spend aggregation.
- End-to-end pipeline completion without error.

Risks & Mitigations

- Ambiguity in SCD Type 2 attribute fields; mitigation: seek clarification or define during technical design.
- Data quality risk; mitigation: rely on cleansing rules as defined.
- Absence of security, error handling, and monitoring detail; mitigation: to be addressed in technical phase.

Approval

- Approval details not present in requirements.

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
customerdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded into Delta table customer
orderdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded into Delta table order

Appendix B: Target Definitions

Layer	Table	Columns	Notes
Bronze	customer	CustId, Name, EmailId, Region	Loaded from customer
Bronze	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId, TotalAmount	Loaded from order
Silver	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	Populated via join, SCD
Gold	customeraggregatespend	Name, TotalAmount, Date	Aggregated spend

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Remove Null literals	customer	customer cleaned	Remove "Null"/Null from all columns
2	Remove duplicates	customer	customer deduped	Drop duplicate rows
3	Remove Null literals	order	order cleaned	Remove "Null"/Null from all columns
4	Remove duplicates	order	order deduped	Drop duplicate rows
5	Add computed column	order	order with TotalAmount	PricePerUnit × Qty
6	Join and SCD Type 2	customer, order	ordersummary	Join on CustId, apply SCD Type 2 rules
7	Aggregate calculation	ordersummary	customeraggregatespend	Group by Name, Date; sum TotalAmount

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-01	Computation	TotalAmount = PricePerUnit × Qty	Requirement
BR-02	Cleansing	Remove "Null"/Null literals and duplicates in all tables	Requirement
BR-03	Join	Join customer and order on CustId	Requirement
BR-04	SCD Type 2	Close previous, open new records, set flags, maintain history	Requirement
BR-05	Aggregation	Group by Name, Date, sum TotalAmount for spend table	Requirement

Appendix E: Process Flow

Step	Description	Dependency
1	Ingest customer and order CSVs as Delta tables	Source files

2	Cleanse customer and order data	Step 1
3	Compute TotalAmount on order	Step 2
4	Join and SCD Type 2 handling for ordersummary	Step 3
5	Aggregate customeraggregatespend table	Step 4

Process Flow Diagram

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